Ebb — Contraction Tracker: Implementation Spec

Oct 25, 2026 · @Marco

How to use this spec

Build Ebb, an offline-first, installable PWA for timing contractions, exactly as drawn in the screenshots below. Everything is stored on the device, with no accounts, no network calls after install and no analytics. The app records and describes; it never diagnoses or says whether someone is in labour.

For Claude Code:

- Treat the screenshots and the Design tokens section as the visual source of truth. The canvas source files (`.dc.html`) are reference HTML: copy their spacing, radii and copy, not their runtime.
- Treat every acceptance criterion (AC) as a test to write. ACs are numbered by screen, e.g. **LT-04**.
- When this spec and the screenshots disagree, the spec wins for behaviour and the screenshots win for layout.
- Store raw timestamps only. Every number on screen is derived (Data model section).
- Never add red, alarms, countdowns or medical labels. The Copy rules section is a hard constraint, not a style preference.

Users. The labouring person taps start/stop one-handed, in pain, often in the dark. The birth partner usually holds the phone, reads numbers out to a midwife and edits entries. The phone passes between them, and a session can run for 12+ hours with the screen on.

In scope for v1: live tracking, intensity, events, sessions, history, session detail with charts, comparison, user thresholds, share text, PDF/CSV export, JSON backup and restore, settings.

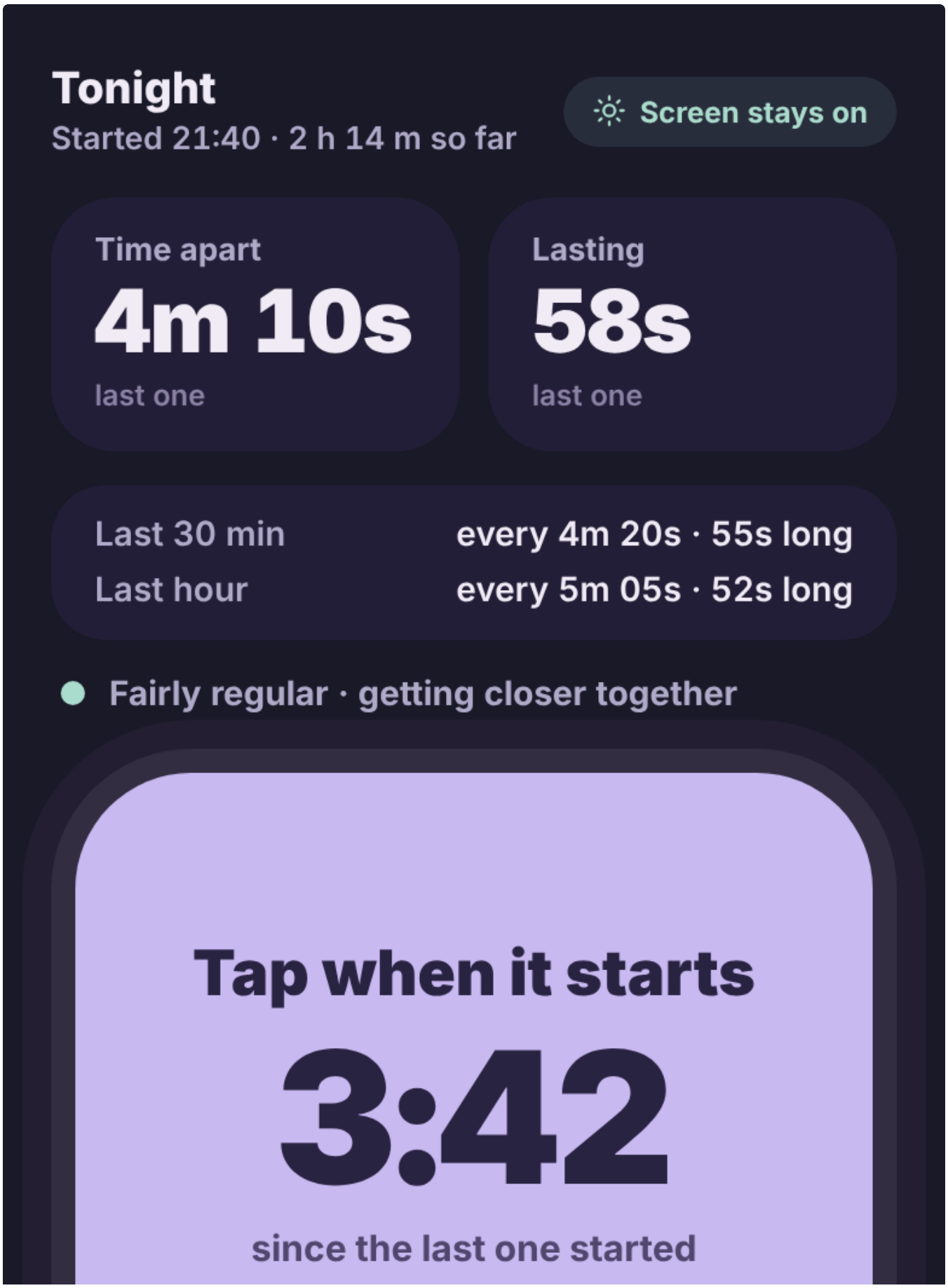
Out of scope for v1: accounts, cloud or partner sync, push notifications, clinical advice, pregnancy tracking, kick counting, hospital bag lists.

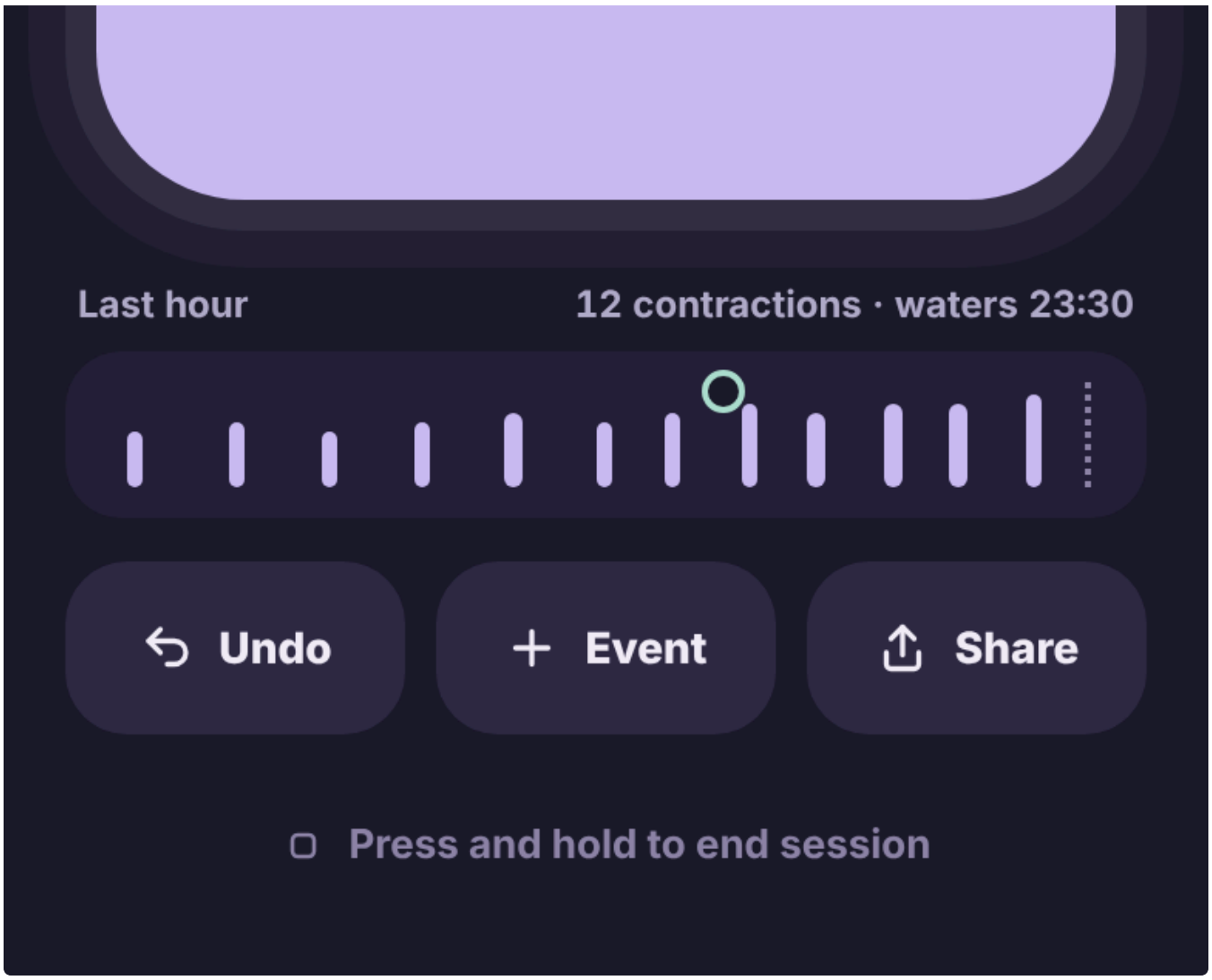
Screens

The app has six screen types. The live tracker has three states, giving eight frames, all at 390 × 844 pt (iPhone 14/15 portrait). The editable source is the Design canvas.

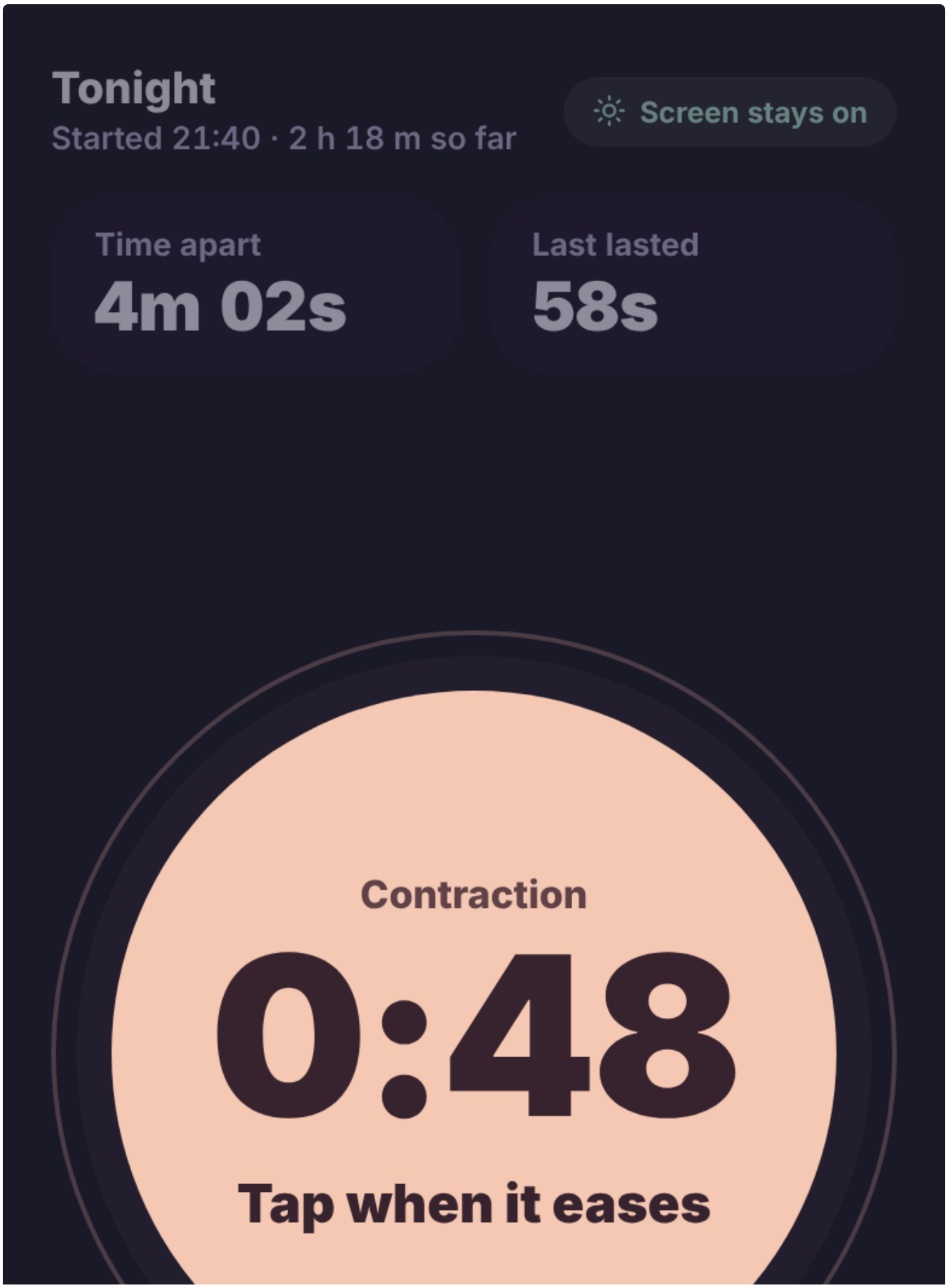
Font note: these screenshots were rendered with Inter as a stand-in because Nunito wasn't available offline. **The app must use Nunito** (Design tokens section). Expect Nunito to look slightly rounder and narrower.

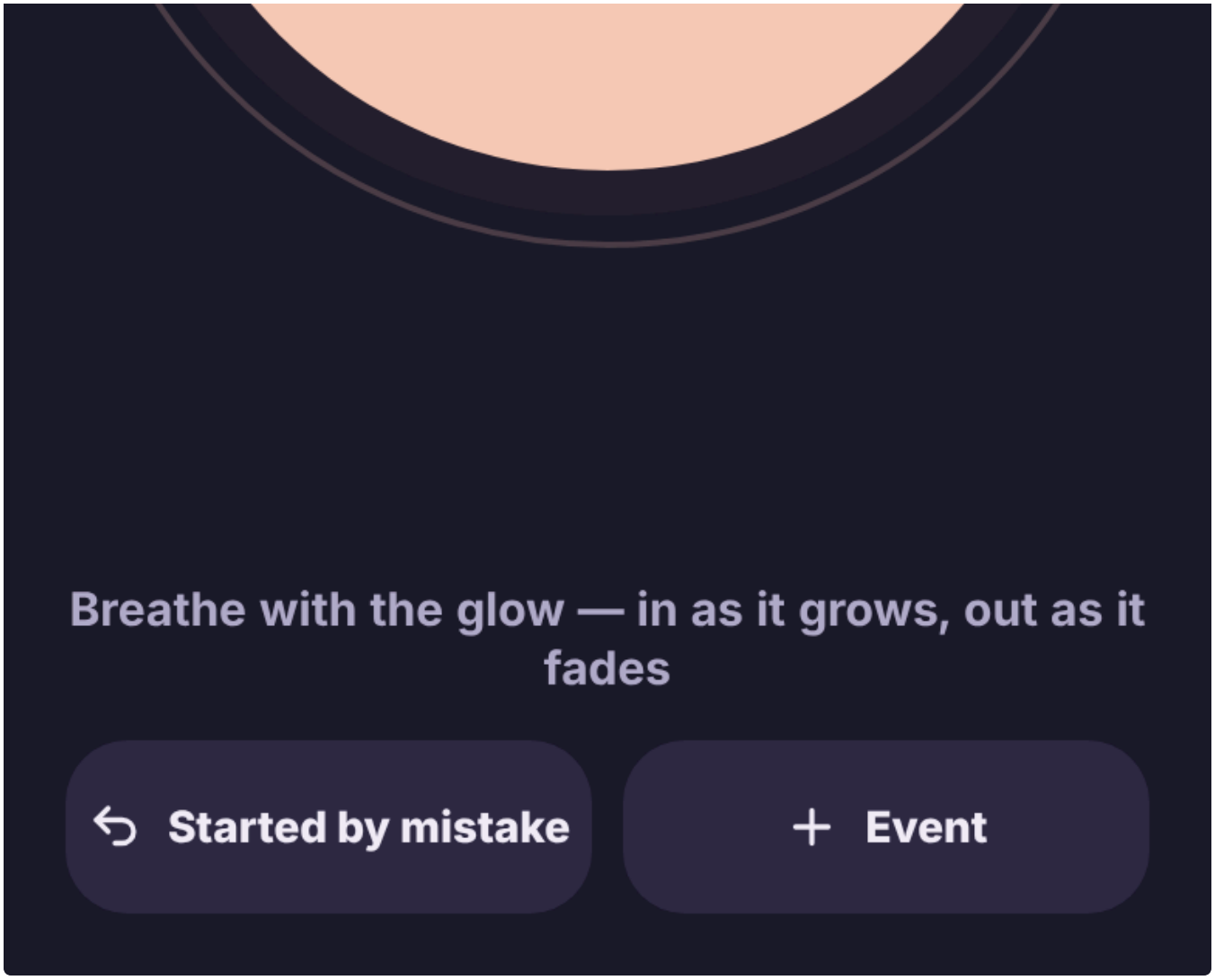
1. Live tracker: resting (ID LT)


Tonight
Started 21:40 · 2 h 14 m so far
Screen stays on
Time apart
4m 10s
last one
Lasting
58s
last one
Last 30 min
every 4m 20s · 55s long
Last hour
every 5m 05s · 52s long
Fairly regular · getting closer together
Tap when it starts
3:42
since the last one started



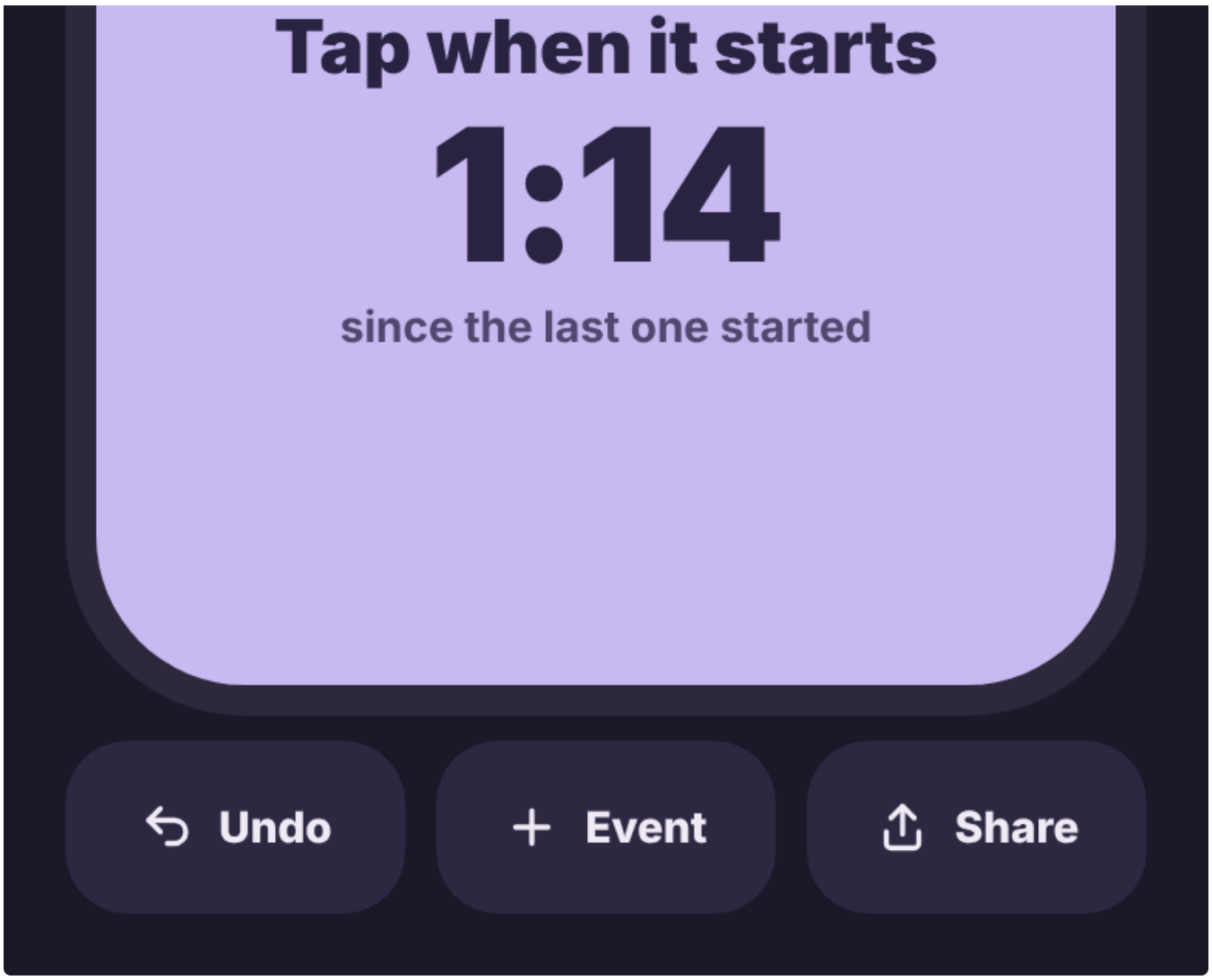
2. Live tracker: during a contraction (ID LT)


Tonight
Started 21:40 · 2 h 18 m so far
Screen stays on
Time apart
4m 02s
Last lasted
58s
Contraction
0:48
Tap when it eases

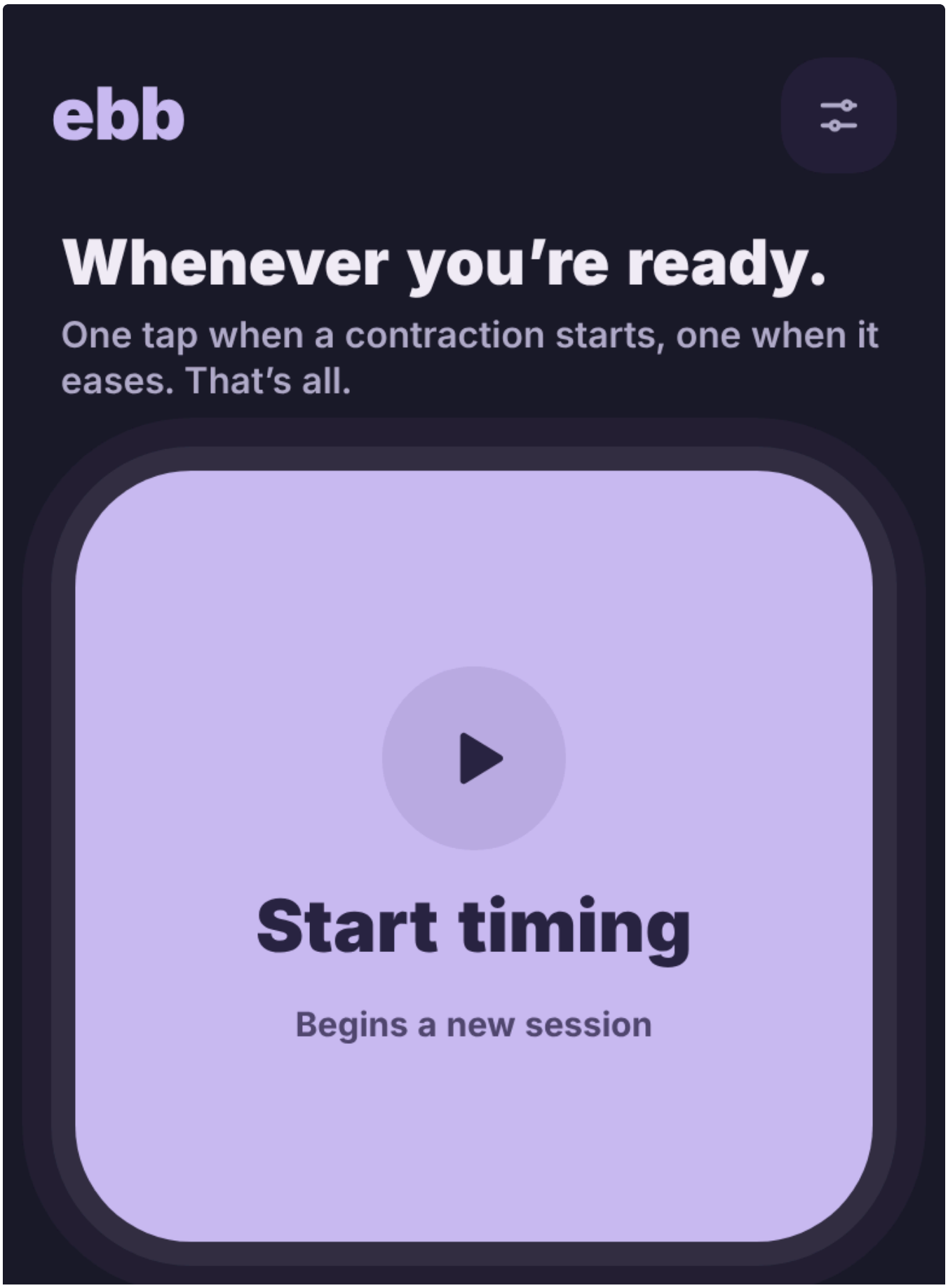


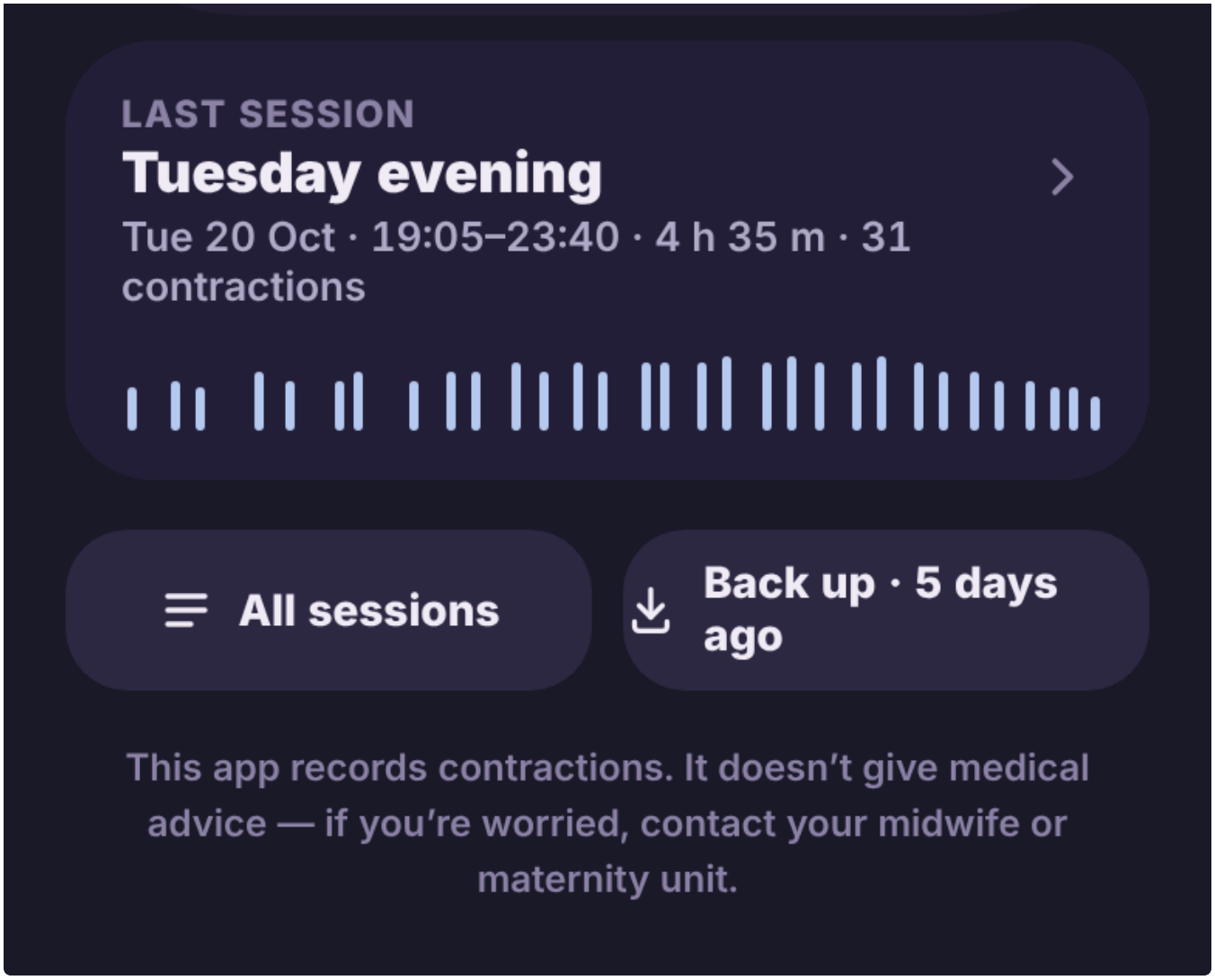
3. Live tracker: just ended, rate it (ID LT)


Tonight
Started 21:40 · 2 h 19 m so far
Screen stays on
Matches the threshold you set
3 in 10 min — from your plan with the midwife
How strong was that one?
1m 02s
Mild
Medium
Strong
Very strong
Intense
Closes by itself — you can rate it later
Skip

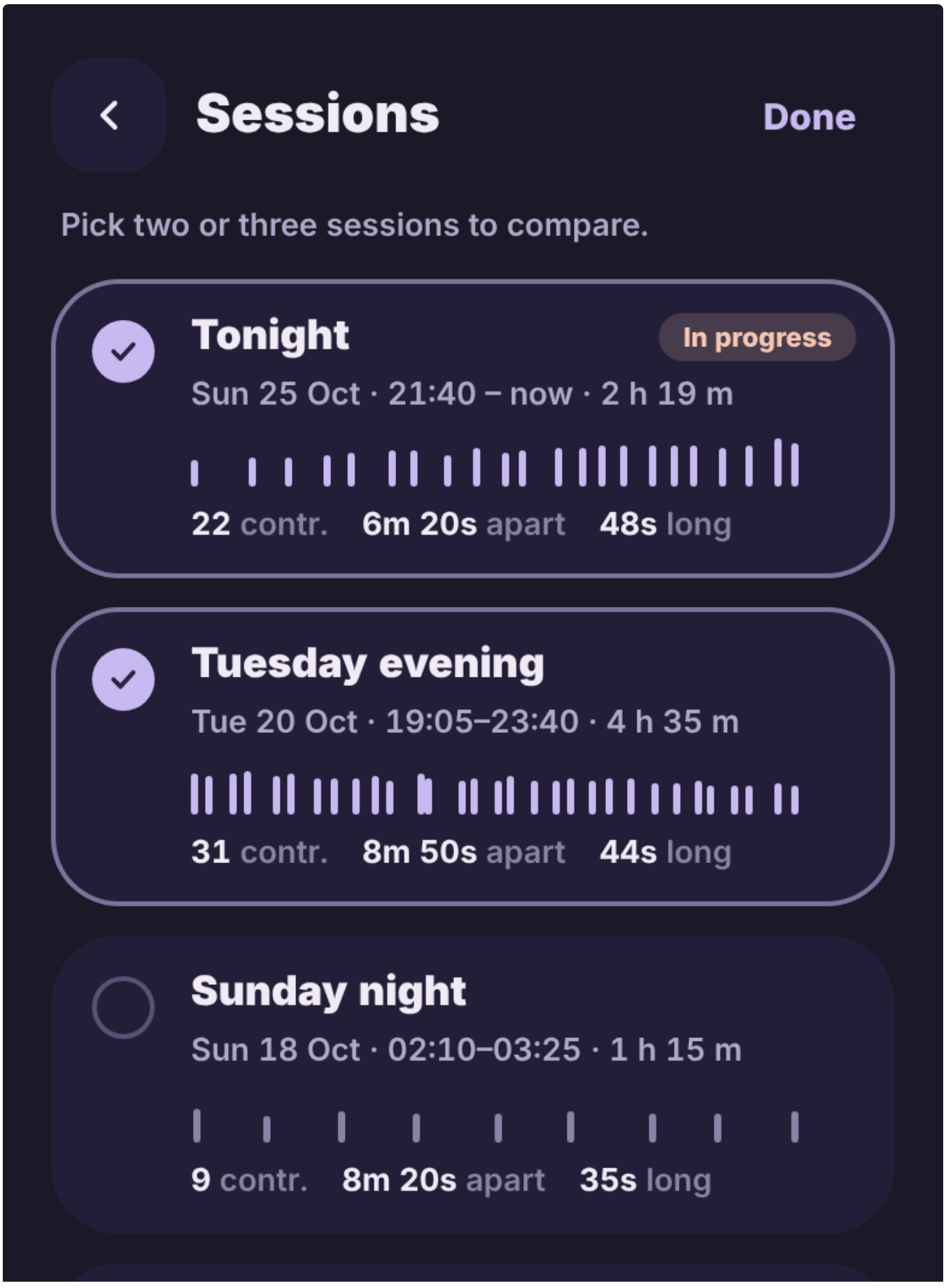


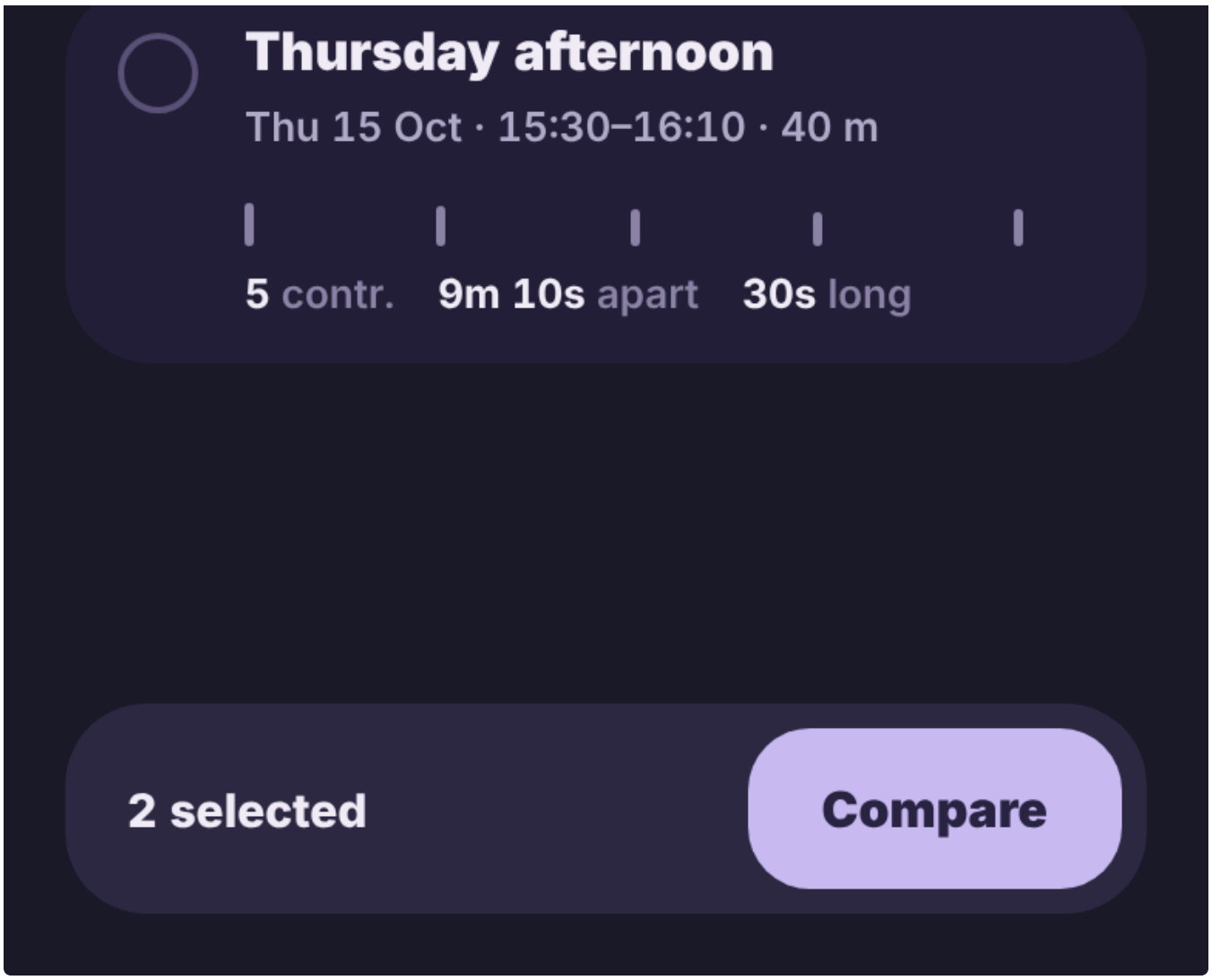
4. Start, no active session (ID ST)


ebb
Whenever you’re ready.
One tap when a contraction starts, one when it eases. That’s all.
Start timing
Begins a new session



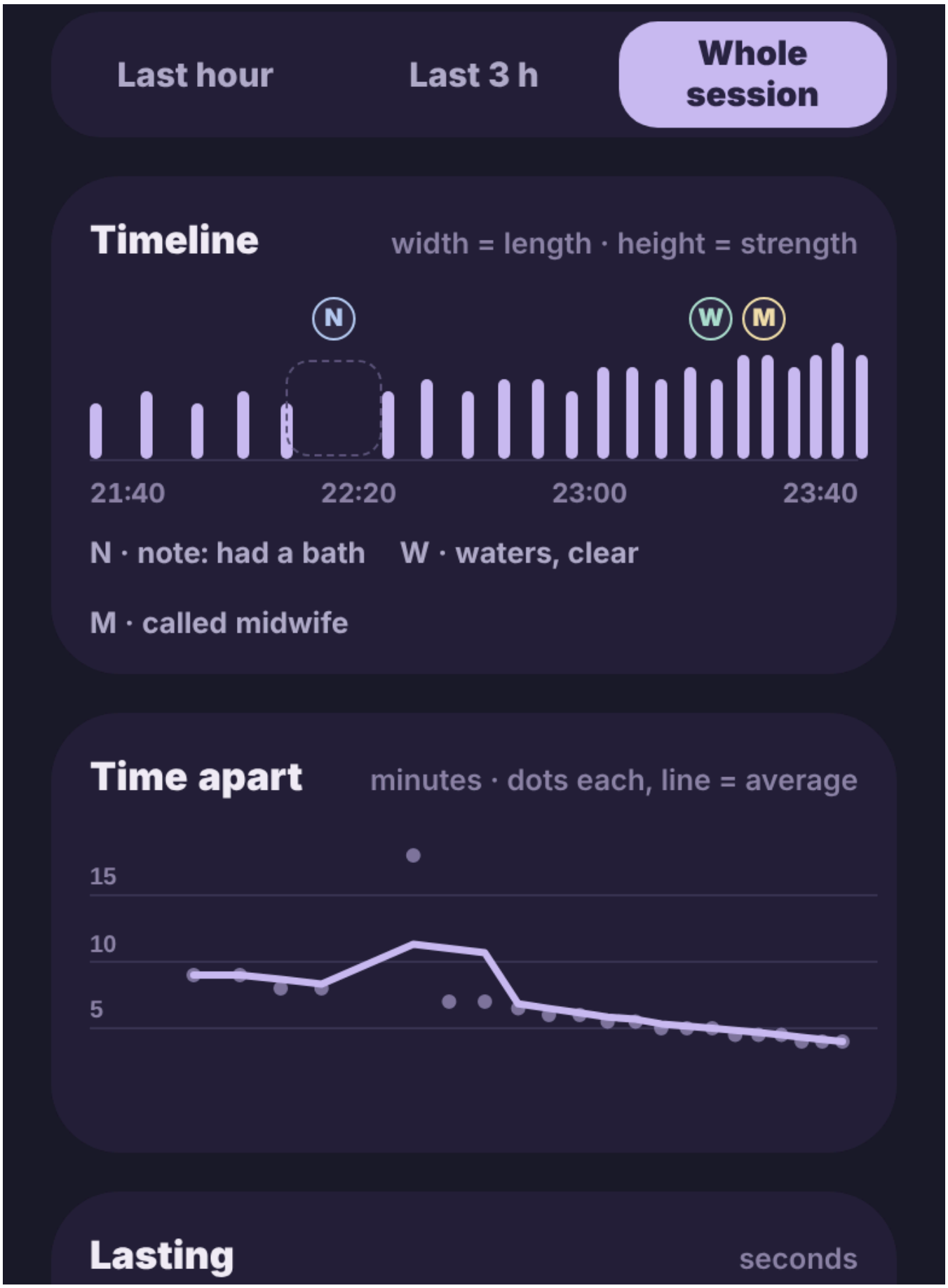
5. History, in select-to-compare mode (ID HI)

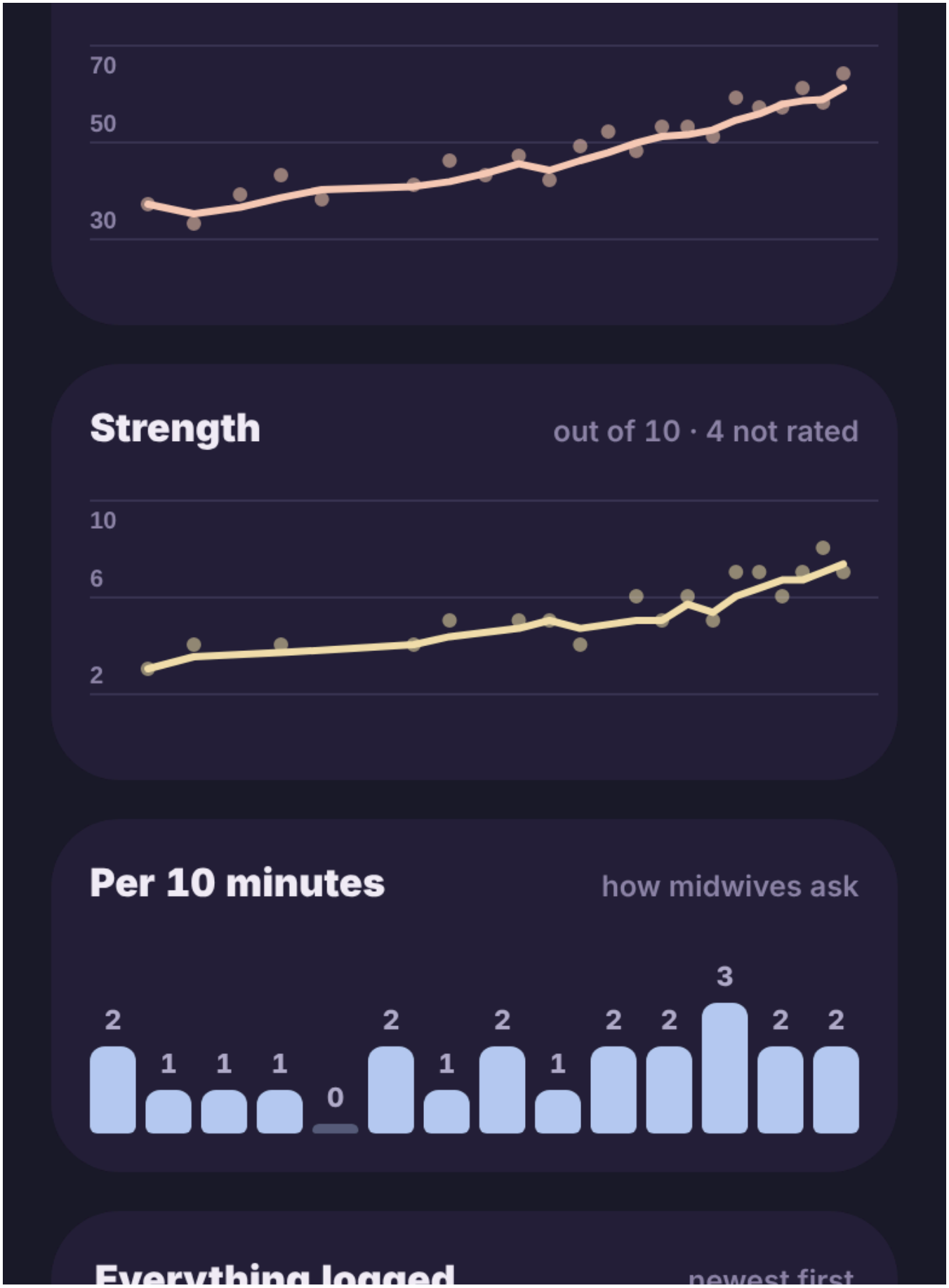

Sessions
Done
Pick two or three sessions to compare.
Tonight
In progress
Sun 25 Oct · 21:40 – now · 2 h 19 m
22 contr. 6m 20s apart 48s long
Tuesday evening
Tue 20 Oct · 19:05–23:40 · 4 h 35 m
31 contr. 8m 50s apart 44s long
Sunday night
Sun 18 Oct · 02:10–03:25 · 1 h 15 m
9 contr. 8m 20s apart 35s long

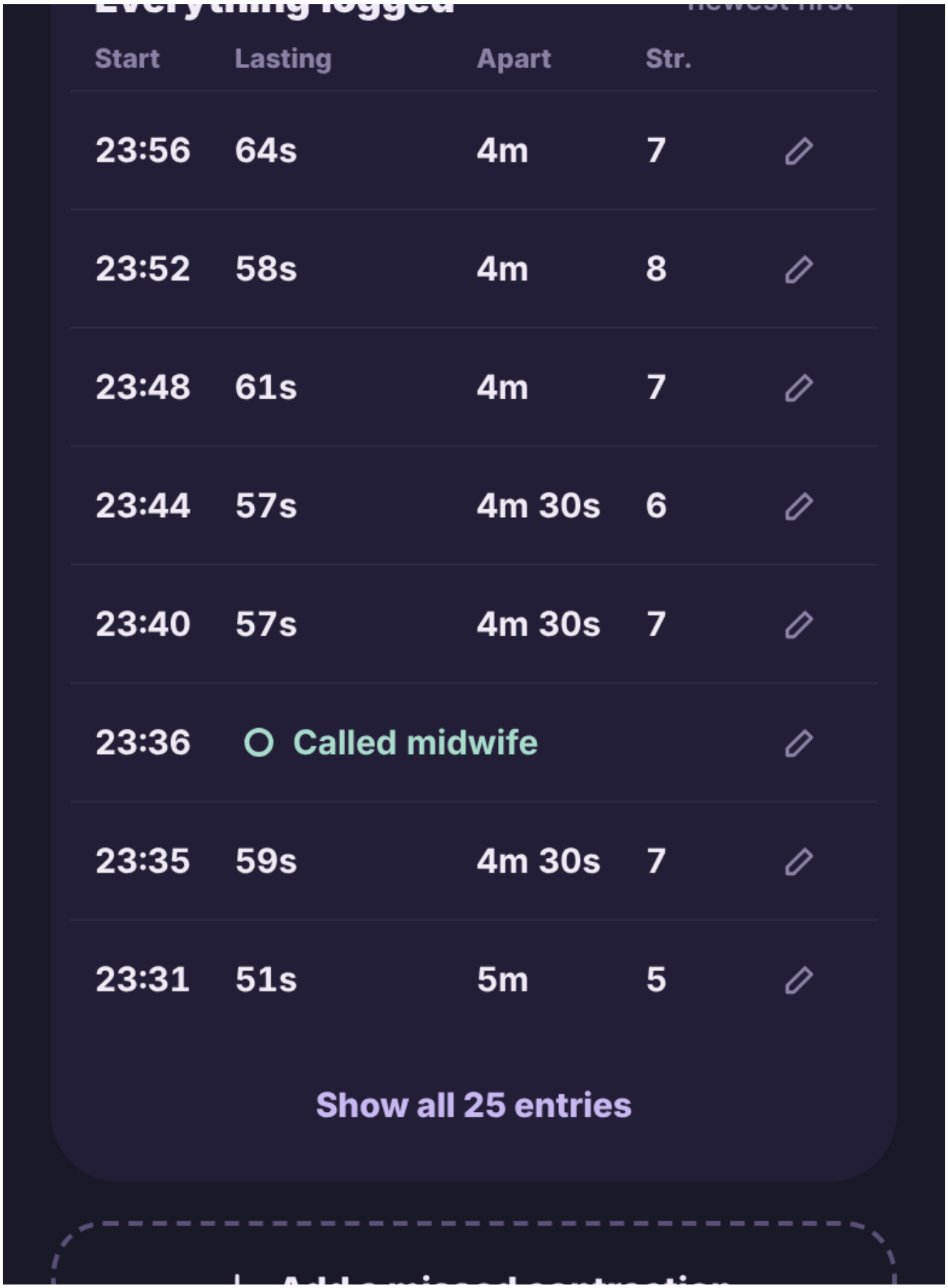


6. Session detail, full scroll (ID SD)


Tonight
Sun 25 Oct · 21:40 – now · 2 h 19 m
Contractions
22
12 in the last hour
Time apart
4m 40s
avg, last hour
Lasting
56s
avg, last hour
Strength
6/10
avg, last hour
Fairly regular
Every 4–6 min over the last hour
Time apart
getting closer together
Length
getting longer
Strength
getting stronger

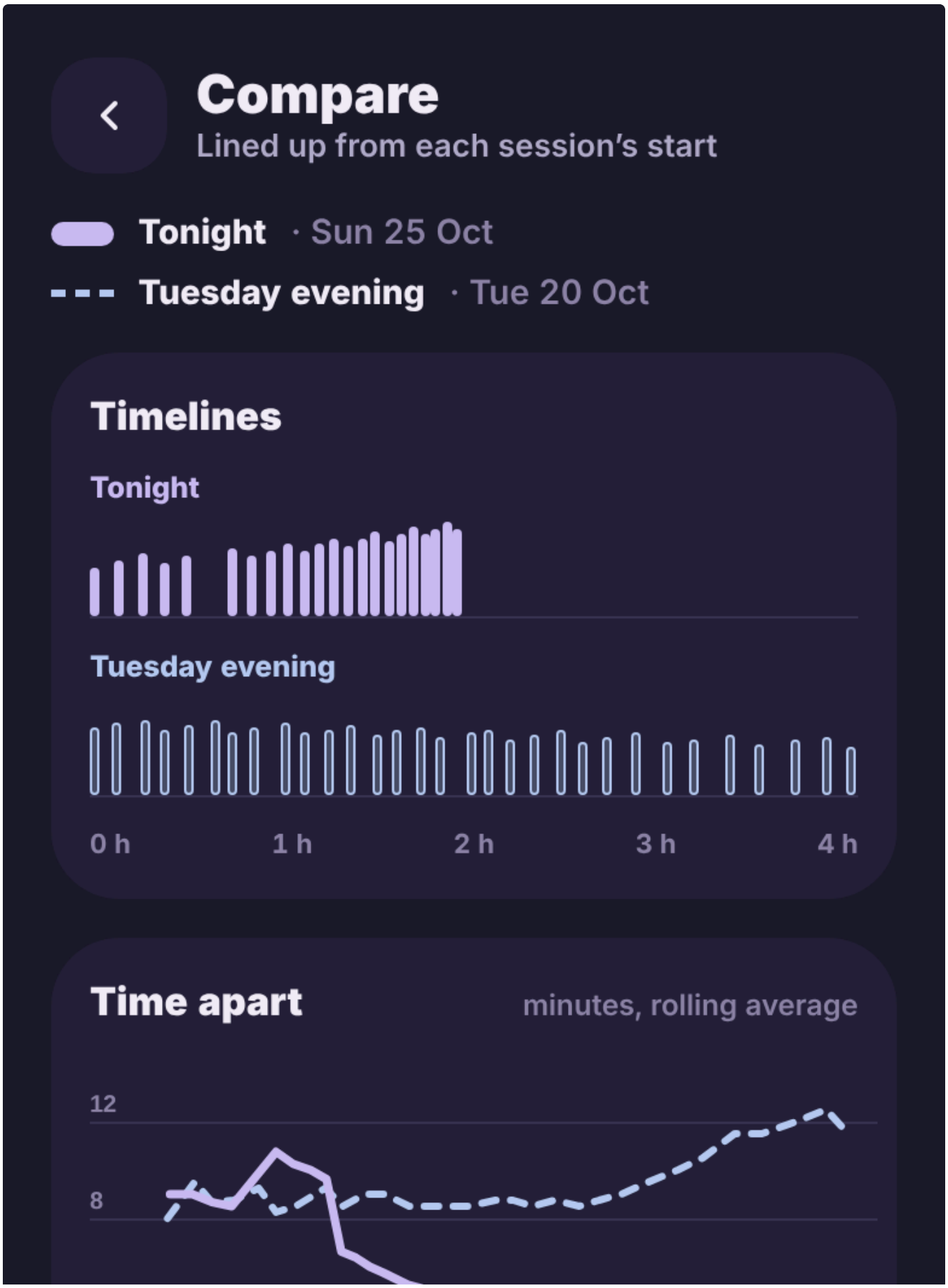

Last hour
Last 3 h
Whole session
Timeline
width = length · height = strength
N
W
M
21:40
22:20
23:00
23:40
N · note: had a bath
W · waters, clear
M · called midwife
Time apart
minutes · dots each, line = average
15
10
5
Lasting
seconds

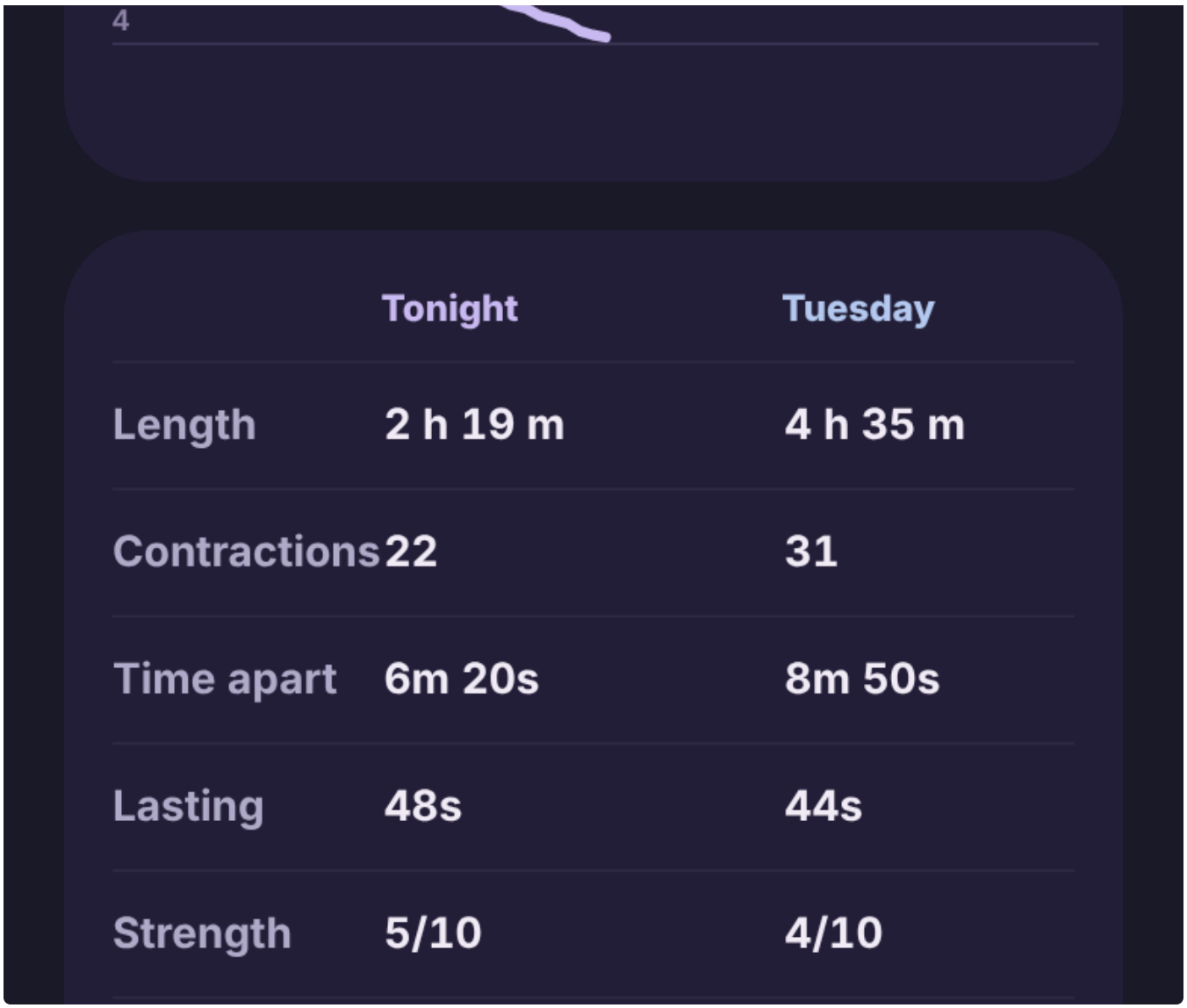

70
50
30
Strength
out of 10 · 4 not rated
10
6
2
Per 10 minutes
how midwives ask
2
1
1
1
0
2
1
2
1
2
2
3
2
2
Everything logged
newest first


Everything logged
newest first
Start
Lasting
Apart
Str.
23:56 64s 4m 7
23:52 58s 4m 8
23:48 61s 4m 7
23:44 57s 4m 30s 6
23:40 57s 4m 30s 7
23:36 Called midwife
23:35 59s 4m 30s 7
23:31 51s 5m 5
Show all 25 entries

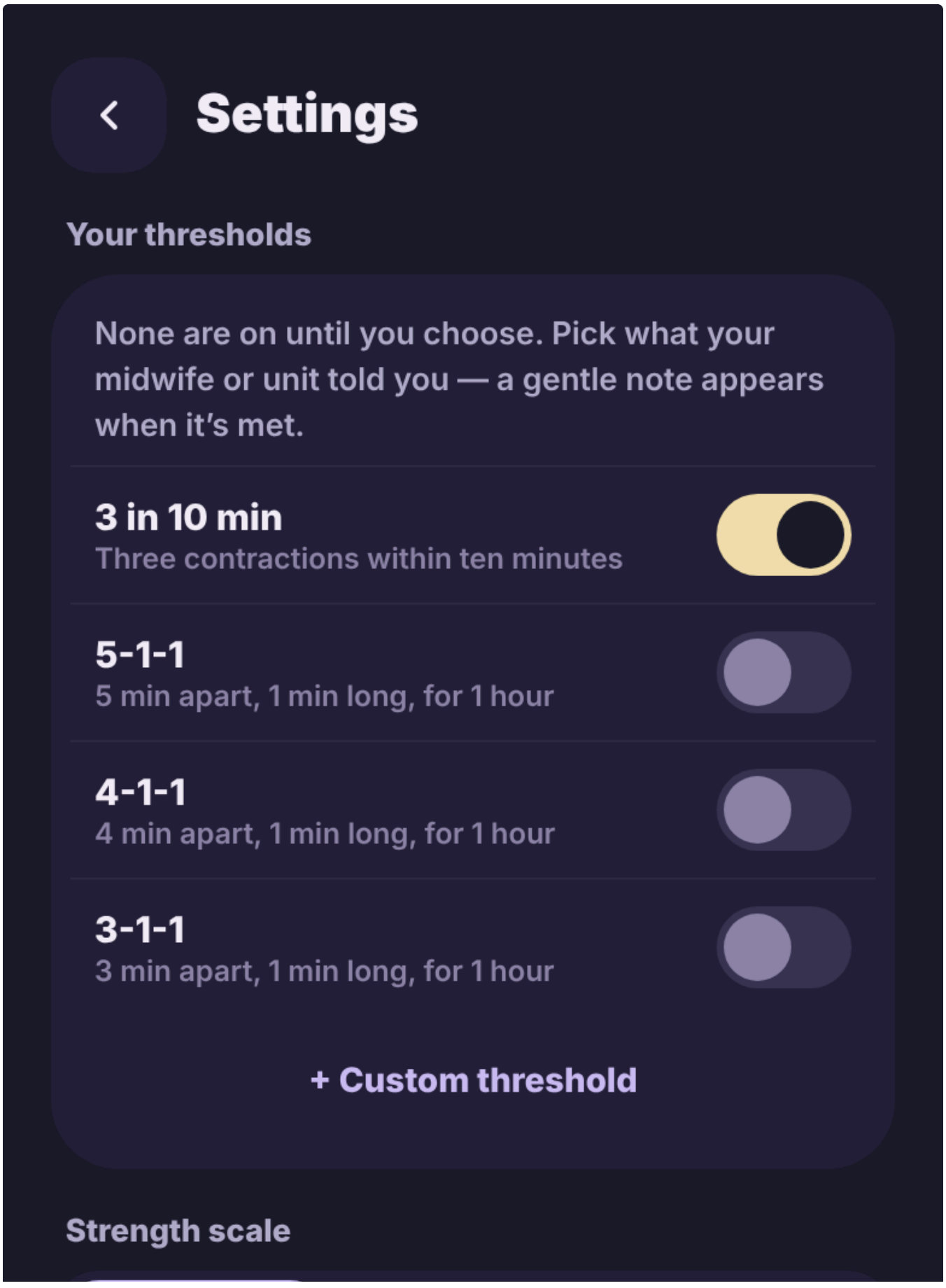


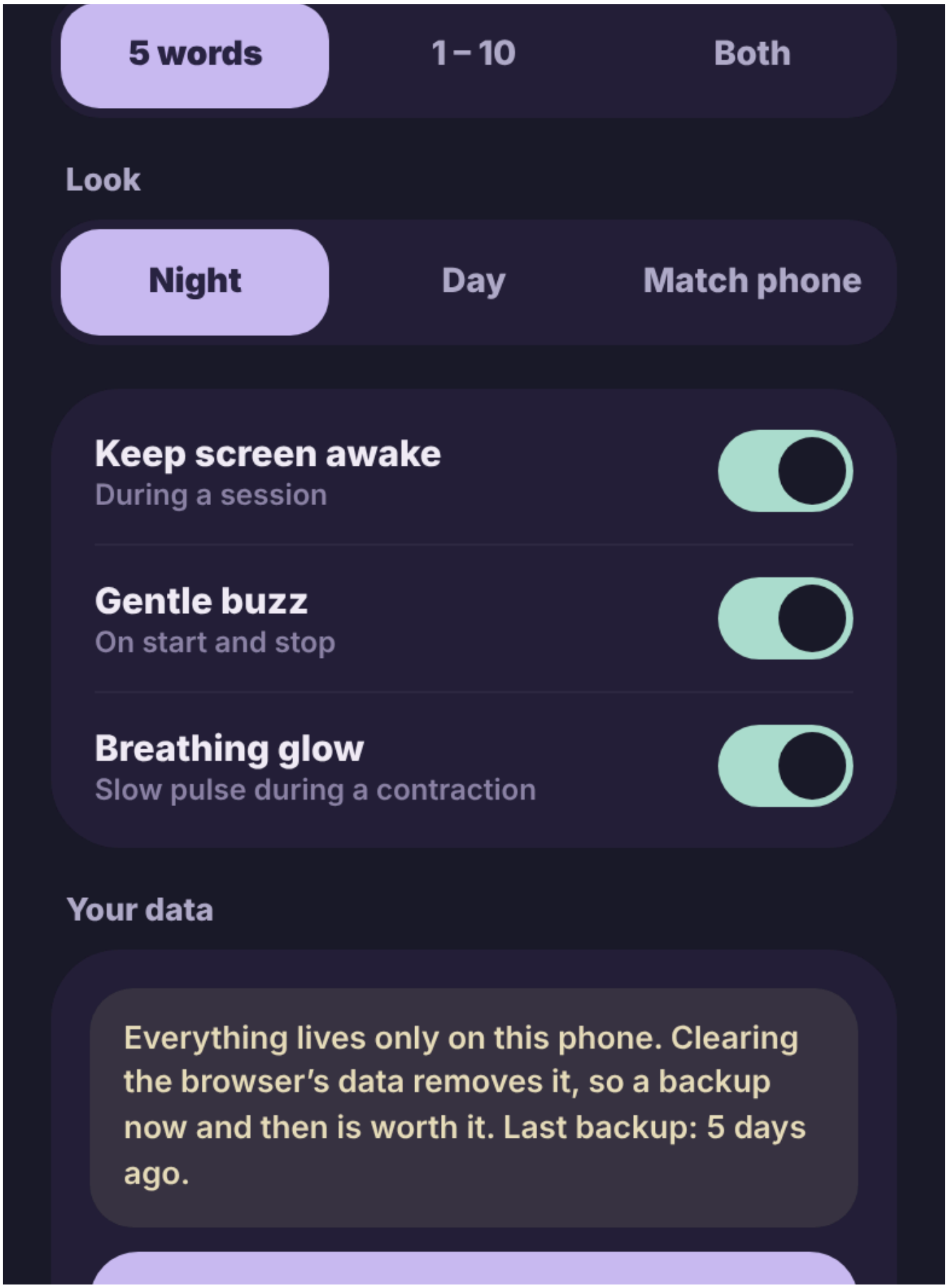
7. Compare sessions (ID CO)

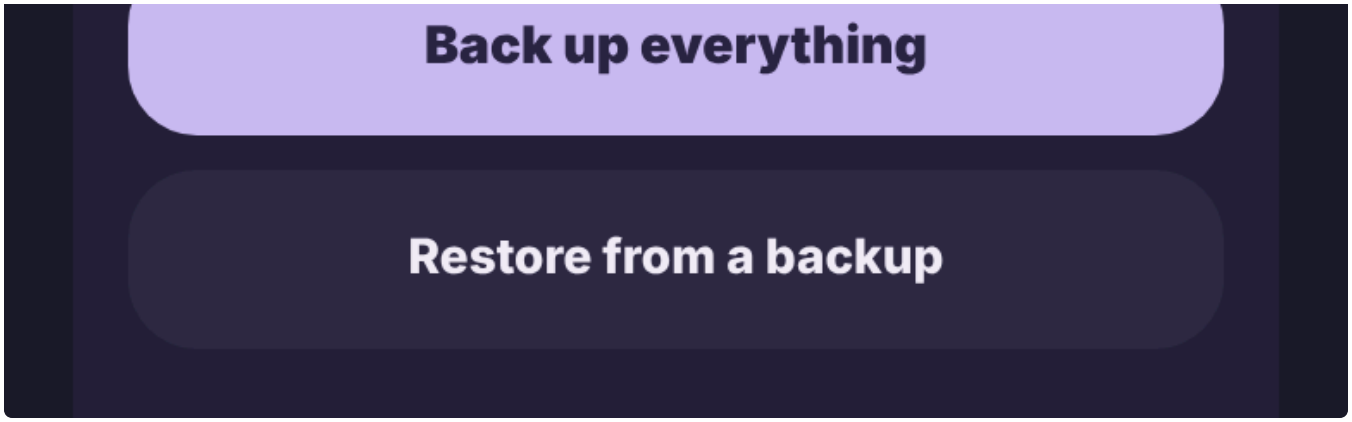

Compare
Lined up from each session's start
Tonight · Sun 25 Oct
Tuesday evening · Tue 20 Oct
Timelines
Tonight
Tuesday evening
0 h
1 h
2 h
3 h
4 h
Time apart
minutes, rolling average
12
8



8. Settings (ID SE)


Settings
Your thresholds
None are on until you choose. Pick what your midwife or unit told you — a gentle note appears when it's met.
3 in 10 min
Three contractions within ten minutes
5-1-1
5 min apart, 1 min long, for 1 hour
4-1-1
4 min apart, 1 min long, for 1 hour
3-1-1
3 min apart, 1 min long, for 1 hour
+ Custom threshold
Strength scale


5 words
1 – 10
Both
Look
Night
Day
Match phone
Keep screen awake
During a session
Gentle buzz
On start and stop
Breathing glow
Slow pulse during a contraction
Your data
Everything lives only on this phone. Clearing the browser’s data removes it, so a backup now and then is worth it. Last backup: 5 days ago.



Palette, type and shapes reference

ebb

Working name. A dusk-plum night, pastel light, nothing sharp.

NIGHT GROUNDS

Dusk
#1B1929 · page

Cushion
#252238 · cards

Pillow
#2F2B45 · buttons

Moonlight
#F1ECF8 · text

PASTELS — EACH HAS ONE JOB

Lavender
#CBBBF4 resting, ready

Peach
#F7C9B5 contraction on

Mint
#AEDFCF events, OK

Butter
#F1E0AA your thresholds

Sky
#B4CAF0 2nd series

No red, no white, no flashing. Text on a pastel is always deep ink #2A2442. Intensity is shown by size or height, never colour alone.

TYPE — NUNITO, ROUNDED TERMINALS, TABULAR FIGURES

3:42 **4m 10s** **Card title** Body 14–15

SHAPES — RADIUS GROWS WITH SIZE

48 · hero

26 · card

20 · button

pill

Design tokens

Put every value below into one tokens file (CSS custom properties) and use nothing outside it. The aim is calm: a dusk-plum night background, pastel accents that each mean one thing, and no pure white, red or flashing.

Typeface: Nunito

- **Family:** Nunito (Google Fonts, SIL Open Font License). Chosen for its rounded terminals, which read as soft, and its clear figures.
- **Self-host it.** Bundle the woff2 files for weights **600, 700, 800 and 900** with the app and precache them in the service worker. Don't load from Google Fonts at runtime, because the app must work offline from first load.
- **Fallback stack:** `'Nunito', ui-rounded, 'SF Pro Rounded', system-ui, sans-serif`.
- **Every number** uses `font-variant-numeric: tabular-nums` so timers don't jitter as digits change.
- Use `font-display: block` for at most 300 ms, then swap, so the timer never flashes in the fallback font on reopen.

Role	Size / weight	Used for
Hero timer	76–92 px / 900, letter-spacing −1 to −2 px	Running timer in the big button
Stat value	36–40 px / 900	Time apart, Lasting
Button label	22–30 px / 900	Tap when it starts, Tap when it eases
Screen title	22 px / 900	Sessions, Settings
Card title	16–19 px / 900	How strong was that one?, chart titles
Body	14–15 px / 600–700	Rows, descriptions
Caption	12–13 px / 600–800	Labels, axis text, disclaimers

Keep the minimum at 12 px. On the live screen the main figures (timer, last interval, last duration) must stay readable from 1 m away.

Colour

Token	Hex	Role
`--bg` (Dusk)	#1B1929	Page background
`--surface` (Cushion)	#252238	Cards
`--surface-2` (Pillow)	#2F2B45	Secondary buttons, table dividers
`--hairline`	#3A3553	Grid lines, inactive tracks
`--text` (Moonlight)	#F1ECF8	Primary text
`--text-muted`	#B3ACCB	Labels
`--text-faint`	#8C85A8	Captions (passes 4.5:1 on `--bg`)
`--ink`	#2A2442	Text on any pastel
`--lavender`	#CBBBF4	Ready/resting state, primary actions, contraction bars
`--peach`	#F7C9B5	A contraction is in progress; duration chart
`--mint`	#AEDFCF	Events, "screen stays on", regularity meter
`--butter`	#F1E0AA	User thresholds and their banner; intensity chart
`--sky`	#B4CAF0	Second series (compare), per-10-min bars
`--rose`	#F2B8C6	Destructive text only (Delete all data). Never a fill.

Tinted panels use the accent at 10–16 % alpha on `--surface`, for example the threshold banner `rgba(241,224,170,0.12)`.

Shape, spacing and motion

- **Radius grows with size:** hero button 48, cards 26–30, buttons 20, small icon buttons 18, pills 999, circular contraction button 50 %.
- **Spacing:** 4-pt base; the screen gutter is 20 px, gaps between cards 10–16 px.
- **Touch targets:** at least 48 × 48 px; the main button fills all spare vertical space (at least 240 px tall).
- **Halo:** the hero button has two soft rings in its own colour, at 10 % (10 px) and 5 % (22 px). No drop shadows, no gradients.

- **Breathing glow:** an 8 s ease-in-out loop that scales 0.94 → 1.04 at 35 → 90 % opacity behind the contraction button. It is disabled under `prefers-reduced-motion` and by the Settings toggle.
- **Transitions:** 200–250 ms ease-out colour crossfade between resting and contracting. No bounces, shakes or flashes anywhere.
- **Icons:** 2–2.4 px stroke, rounded caps and joins (Lucide is a good match). No emoji.

Data model and derived metrics

Store raw timestamps in IndexedDB and compute everything else. Write each tap to disk before updating the UI, so a crash or a lock never loses a tap. All times are epoch milliseconds (UTC); display uses the device's local time zone.

Stores (IndexedDB database `ebb`, version 1)

Store	Key	Fields	Indexes
`sessions`	`id` (UUID v4)	`name?`, `startedAt`, `endedAt?` (null = active), `notes`, `createdAt`, `updatedAt`	`startedAt`, `endedAt`
`contractions`	`id`	`sessionId`, `startAt`, `endAt?` (null = in progress), `intensity?` (1–10), `note?`, `deletedAt?`	`sessionId`, `[sessionId+startAt]`
`events`	`id`	`sessionId`, `at`, `type`, `details?`, `deletedAt?`	`[sessionId+at]`
`settings`	`'singleton'`	`thresholds[]`, `intensityScale` (`words` | `numeric` | `both`), `theme` (`night` | `day` | `system`), `keepAwake`, `haptics`, `breathingGlow`, `lastBackupAt?`, `onboardedAt?`	—
`undo`	`'singleton'`	the last reversible action (type + snapshot)	—

- **Event types:** `waters` (details: `clear` | `pink` | `green-brown` | `unsure`), `bleeding`, `reduced-movement`, `called-midwife`, `went-to-hospital`, `note`.
- **Invariants:** at most one session with `endedAt = null`; at most one contraction with `endAt = null`, and only in the active session.
- **Soft delete:** deletes set `deletedAt`, so Undo can restore them; queries filter these out. Hard-delete soft-deleted rows when their session ends plus 24 h.
- **Intensity:** stored as 1–10 whatever scale is shown. The five-word scale maps Mild 2, Medium 4, Strong 6, Very strong 8, Intense 10.

Derived metrics

For contractions $c_1 \ldots c_n$ in a session, ordered by `startAt`, counting only finished, non-deleted ones:

```
\text{duration}_i = \text{end}_i - \text{start}_i \qquad \text{interval}_i = \text{start}_i - \text{start}_{i-1} \qquad \text{rest}_i = \text{start}_i - \text{end}_{i-1}
```

- **Time apart** (UI label) = interval. **Rest** is shown only in session detail.
- **Rolling averages** over the last 30 and 60 min: the mean of intervals whose start falls in the window. Show — until there are at least 2 intervals.
- **Per 10 min:** the count of contraction starts in [now − 10 min, now); charts use fixed 10-min buckets from the session start.
- **Regularity** = the coefficient of variation of intervals in the last 60 min, needing at least 4 intervals. CV < 0.20 → Regular; 0.20–0.40 → Fairly regular; > 0.40 → Irregular. Always show the range too ("every 4–6 min", the 10th–90th percentile, rounded to whole minutes).
- **Progression:** the least-squares slope of interval, duration and intensity against start time over the last 60 min (at least 5 points). Interval slope < −0.5 min/h → "getting closer together"; > +0.5 → "getting further apart"; else "steady". Duration slope beyond ±5 s/h → "getting longer" or "getting shorter". Intensity slope beyond ±0.5/h → "getting stronger" or "easing off".
- **Gap:** an interval more than 2.5 × the median of the previous 5, and at least 15 min. Draw it as a dashed empty box on the timeline. Exclude it from regularity and progression.
- **Formatting:** under 60 s → `58s`; under 60 min → `4m 10s`; otherwise `1 h 15 m`. Timers on buttons use `m:ss` (and `h:mm:ss` past 1 h).

App lifecycle: open, close, resume

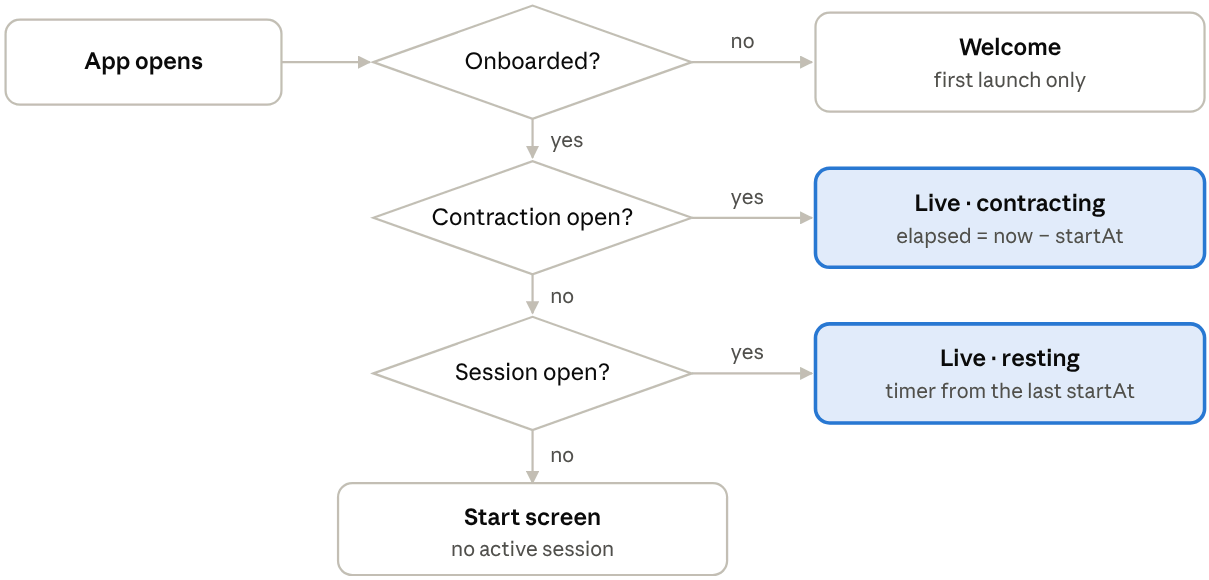
There is no in-app loading screen. On a cold start the OS shows the manifest splash (Dusk background, lavender "ebb" mark). The app shell then paints from the service-worker cache in under 500 ms and lands directly on the right screen. The rules below decide which screen that is.

Where the app lands on open

1. **No `onboardedAt`** → Welcome (first launch, below).

2. **A contraction with `endAt = null`** → Live tracker, contracting state, with elapsed time = now − `startAt`.
3. **A session with `endedAt = null`** → Live tracker, resting state, with "since the last one started" computed from the last `startAt`.
4. **Otherwise** → Start screen.

Ebb always reopens on the screen that matches the stored data



Closing or locking saves nothing extra: every tap is already on disk.

landing rules on every open · 3 checks, 4 destinations

If the IndexedDB read takes longer than 300 ms, show a skeleton of the target screen: card outlines plus the big button in `--surface-2`, with no text and no spinner. Never show a blank or white frame. Set `background-color: #1B1929` on `html` in the inline critical CSS so the very first paint is dark.

First launch (Welcome)

This is one screen, not a carousel:

- The "ebb" mark and the line "Time contractions with one tap. Everything stays on this phone."
- The safety line: "This app records contractions. It doesn't give medical advice — if you're worried, contact your midwife or maternity unit."

- **Install hint:** shown only when not running standalone. On iOS: "Tap Share, then Add to Home Screen." On Android/Chromium: an **Install** button using `beforeinstallprompt`.
- **Primary button: Get started.** It calls `navigator.storage.persist()`, sets `onboardedAt` and goes to Start. Thresholds aren't asked for here; they live in Settings.

While a session is active

- **Wake lock:** request `navigator.wakeLock.request('screen')` when a session starts or the app becomes visible with an active session (if Keep screen awake is on). Re-acquire it on every `visibilitychange` to visible. Release it when the session ends.
- **Wake lock pill:** "Screen stays on" in mint while held. If the API is missing or the request fails, it reads "Screen may sleep" in `--text-faint`, with no alert.
- **Timers:** tick with `requestAnimationFrame`, throttled to 1 Hz for display. Always compute from stored timestamps, never from an incrementing counter.
- **Rolling stats** recompute on every tap and every 15 s.

Closing, locking and backgrounding

- Nothing needs saving on close, because every tap is committed before the UI changes. Never rely on `beforeunload` or `pagehide` to persist data.
- If the phone locks or the app is killed mid-contraction, `endAt` stays null. On reopen the contracting state resumes with the true elapsed time (LT-11).
- **Stale contraction:** on reopen, if an in-progress contraction is older than 5 min, the button still shows the elapsed time. Above it appears a quiet card: "Still going? This one's been running 23 min." with two actions, **Ended earlier…** (opens a time picker for the end) and **Keep timing**.
- **Long pause:** on reopen with a resting session and no contraction for 60+ min, show "Contractions paused? Last one 1 h 12 m ago." with **Keep session going** and **End session…**. Never end a session automatically.

Updates (service worker)

- Precache the app shell, fonts and icons at install. Runtime: cache-first for same-origin assets. There are no network requests for data, ever.
- A new version installs in the background but **never activates or reloads during an active session**. When no session is active, it activates on the next cold start. Settings shows "Update ready — restart the app to use it" until then.

- Bump the IndexedDB schema only with forward migrations in `onupgradeneeded`, and take an automatic JSON snapshot into the `backups` cache first.

User journeys

These seven journeys are the end-to-end tests. Each step names the screen and what must happen.

J1. First contraction of the night (labouring person, alone, one hand)

1. Opens Ebb from the home screen and lands on **Start**.
2. Taps **Start timing**. A session is created and **Live** opens in the resting state with "No contractions yet" and the button reading **Tap when it starts**.
3. A contraction begins: one tap anywhere on the button. It turns peach, a soft haptic plays (if on), and the timer counts up from 0:00.
4. It eases: one tap. The button returns to lavender and the strength picker slides in above it.
5. Ignores the picker; it closes by itself after 30 s. Intensity stays empty and can be added later.

J2. Partner takes over and reads numbers to the midwife

1. Partner picks up the phone. Live already shows the last time apart and length in large type.
2. The midwife asks how often. Partner reads "every 4m 20s over the last 30 minutes, lasting about 55 seconds".
3. Taps **Share**. A sheet shows the summary paragraph with **Copy** and **Share…** (Web Share API). Partner reads it aloud or texts it.
4. Taps **Event → Called midwife**. A mint marker appears on the mini timeline.

J3. A mis-tap

1. The person taps start by accident. The button turns peach.
2. Taps **Started by mistake**. The contraction is discarded and the button goes back to resting, with no confirmation.
3. Or, after stopping a contraction they didn't mean to: taps **Undo** within the resting state, which restores the contraction to in-progress with its original start time.

J4. Threshold met

1. Earlier, in **Settings**, the partner switched on **3 in 10 min**, as the midwife advised.
2. After a contraction ends and the rule is satisfied, the butter banner appears: "Matches the threshold you set — 3 in 10 min."
3. The banner stays while the rule holds and can be dismissed with ×. It reappears only if the rule stops holding and then holds again.

J5. Waters break, with a gentle prompt

1. Taps **Event → Waters broke**, then picks a colour (Clear, Pink, Green or brown, Not sure). Time defaults to now and can be changed.
2. The event saves. Selecting **Green or brown**, **Bleeding** or **Reduced baby movement** shows one calm line below the confirmation: "If you haven't already, contact your midwife or maternity unit." It has no colour change, no icon and no countdown.

J6. Fixing the record later

1. Opens the session from the Live header (taps "Tonight") or from History, and lands on **Session detail**.
2. Taps the pencil on a row and edits start, end, strength and note in a bottom sheet with time pickers. Saving recomputes all charts.
3. Taps **Add a missed contraction**, enters start and length, and it slots into order.
4. Renames the session to "Saturday morning" by tapping the title.

J7. Ending, comparing and backing up

1. On Live, **presses and holds** "Press and hold to end session" for 1.5 s. A ring fills under the finger; letting go early cancels.
2. A sheet asks "End this session?" with a summary and **End session** / **Keep going**. Ending sets `endedAt` and returns to Start.
3. Days later, opens **History →** selects Tonight and Tuesday → **Compare**.
4. In **Settings → Back up everything**, a `ebb-backup-2026-10-25.json` file is downloaded or shared. The Start screen's backup button then shows "Back up · today".

Screens and acceptance criteria

Each AC is one testable statement. Routes are hash-based so the app works from any static host: `#/`, `#/live`, `#/history`, `#/session/:id`, `#/compare?ids=a,b`, `#/settings`.

Live tracker (LT): screens 1–3

The layout from top to bottom: the header (session name linking to detail, elapsed time, wake-lock pill), two stat cards, the averages card, the pattern line, the hero button (flex: 1), the mini timeline (linking to detail), the action row, and the end-session control.

- **LT-01** Tapping anywhere on the hero button toggles resting ↔ contracting. A tap commits to IndexedDB before the UI changes, and the UI responds within 100 ms.
- **LT-02** Resting: lavender, labelled "Tap when it starts", with a big timer for time since the last start (or "No contractions yet" with no timer).
- **LT-03** Contracting: a 300 px peach circle labelled "Contraction" above and "Tap when it eases" below, with an elapsed `m:ss` timer. The header and stat cards drop to 55 % opacity.
- **LT-04** Taps within 600 ms of the previous toggle are ignored (debounce) to absorb double taps.
- **LT-05** The stat cards show the last interval ("Time apart") and last duration ("Lasting"), or — when unknown. Values never wrap; they shrink to 32 px if needed.
- **LT-06** The averages card shows the last 30 and 60 min as "every {interval} · {duration} long", or "Not enough yet".
- **LT-07** The pattern line shows regularity · progression in plain words, with a mint dot for Regular, butter for Fairly regular and faint for Irregular. It is hidden when there's too little data.
- **LT-08** When a contraction ends, the strength card appears with five options (Mild… Intense) shown as growing dots plus words, a numeric 1–10 row, or both (per Settings). Choosing one saves and closes it. **Skip** closes it. With no choice it closes after 30 s, with a thin progress bar. It closes immediately if a new contraction starts.
- **LT-09** The mini timeline shows the last 60 min: bar x = start, width = duration (min 5 px), height = intensity (unrated = middle height, hollow outline). Event markers are circles. A dotted line marks now.
- **LT-10 Undo** reverts the last action (start, stop, rating, event, edit) in one tap, with no confirmation. A toast reads "Undone" for 3 s. Undo depth is 1. During a contraction the left action is labelled **Started by mistake**.

- **LT-11** Reopening the app mid-contraction restores the contracting state with the correct elapsed time, within ±1 s.
- **LT-12 Event** opens a bottom sheet with the six types, an optional time (default now) and an optional note. Saving shows a mint marker.
- **LT-13 Share** opens the summary sheet (see SH).
- **LT-14** Ending the session requires a 1.5 s press-and-hold, then a confirmation sheet. It is never a single tap.
- **LT-15** The threshold banner (butter tint) appears only when an enabled user threshold is met. It uses `role="status"` and is dismissible.
- **LT-16** Haptics (`navigator.vibrate(15)` on start, `[10,60,10]` on stop) fire only if enabled and supported. Absence is silent.

Start (ST): screen 4

- **ST-01** "Whenever you're ready." sits above a lavender hero **Start timing** button that creates a session and opens Live.
- **ST-02** The last-session card shows name, date, time range, length, count and a sparkline, and opens Session detail. It is hidden when there is no history.
- **ST-03 All sessions** opens History. The backup button shows the relative time since the last backup ("never" in `--butter` if none and data exists).
- **ST-04** The safety line is always visible at the bottom.

History (HI): screen 5

- **HI-01** Sessions are listed newest first. Each card shows name (or the default name, e.g. "Sunday evening", from the start's weekday and part of day), date and time range, length, count, average time apart, average length and a sparkline. The active session shows the peach "In progress" pill.
- **HI-02** Tapping a card opens detail. **Select** enters select mode, showing checkbox circles and a bottom bar "{n} selected · Compare".
- **HI-03** Compare is enabled for 2–3 selections. A fourth selection is blocked with the hint "Up to three at a time".
- **HI-04** The empty state reads "Sessions you time will appear here" with a Start timing link.

Session detail (SD): screen 6

- **SD-01** The header has back, an editable title (tap → inline input, saves on blur or Enter) and an Export button.
- **SD-02** Summary: contractions (total and last hour), time apart, lasting and strength averages over the last hour (or the whole session if ended).
- **SD-03** The pattern card shows a three-step regularity meter (shape plus label, not colour alone), the range text and three progression rows with icons.
- **SD-04** The range control (Last hour / Last 3 h / Whole session) sets the x-domain of all charts. Pinch-zoom and drag-pan work on the time axis; double-tap resets.
- **SD-05** Charts: Timeline (hero), Time apart (dots plus a rolling average of 3), Lasting, Strength (unrated count in the subtitle), and Per 10 minutes. Tapping a bar or point shows a tooltip with exact times and an **Edit** link.
- **SD-06** The log table is newest first and mixes contractions and events. Each row has an edit pencil; swiping left reveals Delete (soft, undoable). It shows the latest 8 rows plus "Show all {n} entries".
- **SD-07 Add a missed contraction** opens a sheet with start time and length. It is rejected inline if it overlaps another contraction.
- **SD-08** Notes are a free textarea that autosaves after 500 ms idle.
- **SD-09** Actions: **Share summary**, **PDF report** and **CSV data** (see SH).

Compare (CO): screen 7

- **CO-01** The legend distinguishes series by fill and line style (solid lavender vs dashed sky outline), never by colour alone.
- **CO-02** Timelines are stacked and aligned at t = 0 on a shared hour axis sized to the longest session.
- **CO-03** Interval trends are overlaid as a rolling average of 4.
- **CO-04** The metrics table shows length, contractions, time apart, lasting, strength and pattern, one column per session.

Settings (SE): screen 8

- **SE-01** Threshold presets: 3 in 10 min; 5-1-1; 4-1-1; 3-1-1, all **off** by default. **+ Custom threshold** lets the user define either *N in M minutes* or *A-L-D* (apart ⩽ A min, lasting ⩾ L min, for D hours).
- **SE-02** Strength scale (5 words / 1–10 / Both); Look (Night / Day / Match phone).

- **SE-03** Toggles: Keep screen awake, Gentle buzz and Breathing glow, all on by default.
- **SE-04** Data: **Back up everything** (JSON), **Restore from a backup** (file input, preview counts, then Merge or Replace) and **Delete all data…** (rose text, typed confirmation "delete").
- **SE-05** The safety line and the app version appear at the bottom.

Share and export (SH)

- **SH-01** The summary text follows this template: "Last hour: {n} contractions, every {interval}, lasting about {duration}, strength about {i}/10. {events}". Example: "Last hour: 12 contractions, every 4.5 min, lasting about 55 s, strength about 6/10. Waters broke 23:30, clear."
- **SH-02** The PDF is generated on device (e.g. jsPDF plus SVG charts): a summary, the timeline, the trend charts and the full table, A4 portrait, light theme.
- **SH-03** CSV columns: `type,start,end,duration_s,interval_s,intensity,note`, with ISO 8601 local time and offset.
- **SH-04** The backup JSON is `{app:"ebb", schema:1, exportedAt, sessions[], contractions[], events[], settings}`. Import validates the schema version and IDs, and de-duplicates by `id`.

Thresholds and copy rules

The app describes what happened and points back to the user's own plan. It never interprets the pattern medically.

Threshold evaluation

- Evaluate all enabled thresholds after each contraction ends, after an edit, and every 60 s.
- **N in M:** at least N contraction starts in the last M minutes.
- **A-L-D:** every interval ⩽ A min **and** every duration ⩾ L min (±5 s tolerance) across the last D hours, with at least (D × 60 / A) − 1 contractions in that window and no gap.
- When a threshold is met, show the banner "Matches the threshold you set — {name}", with the user's custom label if set. If several are met, show the first in list order. Record the first match time per session as a timeline marker ("Threshold met").

Copy rules (hard constraints)

Do	Never
"every 4–6 min over the last hour"	"You are in active labour" / "early labour"
"getting closer together", "steady", "easing off"	"Braxton Hicks", "false labour", "progressing well"
"Matches the threshold you set"	"Time to go to hospital", "Call now!"
"If you haven't already, contact your midwife or maternity unit."	Red text, warning icons, sirens, countdowns, exclamation marks
"Contractions paused?"	"Labour has stopped"

- **Voice:** warm, plain, brief, second person. Use sentence case. Use UK spelling and 24-hour times by default, following the locale.
- **Safety line** (verbatim, on Welcome, Start and Settings): "This app records contractions. It doesn't give medical advice — if you're worried, contact your midwife or maternity unit."
- **Gentle prompt** (verbatim) after the events bleeding, reduced baby movement, or waters green or brown: "If you haven't already, contact your midwife or maternity unit."

Edge cases

Each row is a required behaviour and needs a test.

Case	Required behaviour
Double tap or palm tap on the hero button	600 ms debounce (LT-04); Undo covers the rest
Contraction under 5 s	Saved, but marked "very short" in the log; excluded from averages until edited
Contraction over 5 min / app reopened mid-contraction much later	"Still going?" card with **Ended earlier…** and **Keep timing** (Lifecycle section)
No contraction for 60+ min	"Contractions paused?" card; never auto-end

Case	Required behaviour
Session past midnight	Times keep counting; the default name uses the start day ("Saturday night"); charts label dates when crossing midnight
Clock or time-zone change mid-session (travel, daylight saving ends 25 Oct)	Store UTC epoch ms; compute durations from timestamps; if a new `startAt` < the previous one (clock moved back), keep insertion order and flag the row "time changed" for review
Edits that overlap another contraction	Reject inline: "Overlaps the one at 23:44"
Edit puts end before start	The save button is disabled with an inline hint
Rating skipped	Intensity null; the chart uses hollow middle-height bars; averages ignore it; "{n} not rated" in the subtitle
Only 1–3 contractions	Stats show —; the pattern line is hidden; charts render what exists without trend lines
Storage full / write fails	Keep the UI state, retry the write 3×, then show a quiet card: "Couldn't save the last tap — free some space, then tap Retry". Never lose the tap from memory while the app is open
Private browsing / storage unavailable	On launch: "This browser won't keep your data. Open Ebb from your home screen instead." Tracking still works in memory, with a visible "Not saved" pill
`persist()` denied	Silent; the backup nudge on Start becomes more prominent after 3 sessions
Ebb open in two tabs or windows	A `BroadcastChannel` keeps them in sync; writes go through a single IndexedDB transaction queue; the last write wins per record
Wake Lock unsupported (older iOS)	The pill reads "Screen may sleep"; Settings explains how to raise auto-lock time

Case	Required behaviour
Phone call or other app in foreground	Nothing happens; timers recompute from timestamps on return
Restore a backup that overlaps existing data	Merge by `id`; on conflict keep the newer `updatedAt`; show the counts added and skipped
Corrupt or wrong-version backup	Reject with "This file isn't an Ebb backup"; change nothing
Delete all data	Typed confirmation; clears IndexedDB and caches the user data only; returns to Welcome
Very long session (24 h, 300+ contractions)	Charts stay at 60 fps; the log is virtualised; computations stay under 16 ms (memoise per session)
Reduced motion	No breathing glow, no slide-ins (fade only), no fill animation on hold-to-end (a text countdown instead)
Large text (200 % zoom / Dynamic Type)	Layout reflows; the stat cards stack; the hero button keeps at least 200 px height and scrolls if needed
Landscape on phone	Supported: the hero button on the left half, stats and timeline on the right

Accessibility, PWA and performance

Accessibility (WCAG 2.2 AA)

- All text is at least 4.5:1 against its background. Pastel fills always carry `--ink` text.
- Use real elements: `<button>`, `<a>`, `<input>`, `<label>`, `<table>` with `<th scope>`. Toggles are `role="switch"` with `aria-checked`; the strength picker is a `radiogroup`.
- The hero button's accessible name includes state and time, e.g. "Start contraction. 3 minutes 42 seconds since the last one started". An `aria-live="polite"` region announces "Contraction started" and "Contraction ended, lasted 58 seconds".

- Charts have `role="img"` with a sentence summary, plus the log table as the accessible equivalent.
- Intensity and series are never encoded by colour alone; size, height, fill style or dashes always back them up.
- Respect `prefers-reduced-motion` and `prefers-contrast: more` (raise `--text-muted` to #D6D0E8).
- Focus rings are 3 px lavender with a 2 px offset and visible on keyboard focus only.

PWA

- **Manifest:** `name` "Ebb — Contraction Timer", `short_name` "Ebb", `display: standalone`, `orientation: any`, `background_color` and `theme_color` #1B1929. Icons at 192, 512 and maskable 512, plus an Apple touch icon at 180.
- **Meta:** `apple-mobile-web-app-capable`, `apple-mobile-web-app-status-bar-style: black-translucent`, and `viewport-fit=cover`, with safe-area insets padding the top and bottom.
- **Service worker:** precache everything at install; the app works in airplane mode after the first load (Lighthouse PWA installable and offline checks pass).
- **Privacy:** zero network requests after install (verify with a test that fails on any `fetch`). No analytics or crash reporting in v1.

Performance budgets

- First paint under 500 ms on cached reopen (mid-range Android); interactive under 1 s.
- JS under 120 kB gzipped, excluding the PDF library, which is lazy-loaded only on export.
- A tap's visual response under 100 ms; DB commit under 50 ms at the 95th percentile.
- Battery: with the screen awake for 8 h, no animation besides the 1 Hz timer and the optional glow, which uses CSS only and pauses when hidden.

Stack, architecture and build order

Use Vite + React + TypeScript + Zustand, with Dexie over IndexedDB, and ship it as a static PWA with no backend.

Concern	Choice	Why
Build	Vite, TypeScript strict	Fast, static output

Concern	Choice	Why
UI	React 18, CSS Modules + a `tokens.css` of custom properties	Tokens map 1:1 to this spec
State	Zustand store derived from Dexie live queries	The DB is the source of truth; the store only caches
Storage	Dexie 4 (IndexedDB)	Indexes, transactions, migrations
PWA	`vite-plugin-pwa` (Workbox `generateSW`, `registerType: 'prompt'`)	Precache and controlled updates
Charts	Hand-rolled SVG components + d3-scale / d3-array only	Small, themable, accessible; pinch-zoom via pointer events
Font	`@fontsource/nunito` 600/700/800/900, latin subset	Self-hosted, precached
Icons	`lucide-react`	Rounded strokes
PDF	`jspdf` + `svg2pdf.js`, dynamically imported	Off the main bundle
Tests	Vitest + `fake-indexeddb` (unit), Playwright with WebKit and Chromium mobile emulation (E2E)	Covers the ACs and journeys

Module layout

- `src/db/`: the Dexie schema, migrations, repositories (sessions, contractions, events, settings) and the undo log.
- `src/domain/`: pure functions (intervals, rolling averages, CV, slopes, gaps, threshold evaluation, formatting, summary text). 100 % unit-tested; no DOM.
- `src/app/`: the router (hash), lifecycle (landing rules, wake lock, visibility) and service-worker update handling.
- `src/screens/`: Live, Start, History, SessionDetail, Compare, Settings, Welcome.
- `src/components/`: HeroButton, StatCard, StrengthPicker, Sheet, Toast, Switch, Segmented, HoldToConfirm.
- `src/charts/`: Timeline, TrendChart, FrequencyBars, Sparkline.

Build order (each milestone ends green on its ACs)

1. **Foundations:** tokens, Nunito, the app shell, manifest, service worker, Dexie schema, hash router, Welcome. Offline test passes.
2. **Live core:** Start → session → HeroButton toggle, timers from timestamps, persist-before-render, resume after reload, debounce. LT-01–04, LT-11.
3. **Live info:** domain metrics, stat cards, averages, pattern line, mini timeline, wake lock pill. LT-05–07, LT-09.
4. **Corrections:** StrengthPicker, Undo / Started by mistake, Event sheet, hold-to-end. LT-08, LT-10, LT-12, LT-14, LT-16.
5. **Looking back:** History, Session detail with all charts, editing, add missed, notes. HI, SD.
6. **Thresholds and settings:** threshold engine + banner, Settings. LT-15, SE.
7. **Sharing and data:** summary text, CSV, PDF, backup/restore, delete all. SH, SE-04.
8. **Compare** and the remaining edge cases (stale contraction, paused, multi-tab, storage errors).
9. **Polish:** accessibility audit (axe + VoiceOver pass), reduced motion, large text, landscape, performance budgets.

Definition of done

- Every AC and journey has a passing test. The seven journeys run as Playwright specs on iPhone 13 (WebKit) and Pixel 7 (Chromium) emulation.
- The visuals match the screenshots side by side at 390 × 844 (Nunito rendering differs slightly from the screenshots, as noted).
- Lighthouse: PWA installable, Accessibility at least 95, Performance at least 90.
- Airplane-mode test: install, go offline, kill the app, reopen and run J1.

Decisions taken and open questions

Question from the brief	Decision for v1
Partner sync	One device. Reserve the header pill slot (where "Screen stays on" sits) for a future "Shared session" indicator
Intensity scale	Five words by default, stored as 2/4/6/8/10; 1–10 and Both are available in Settings

Question from the brief	Decision for v1
Breathing visual	Yes, as the 8 s glow behind the contraction button; on by default, off under reduced motion
Name and identity	Working name "Ebb", lowercase wordmark in Nunito 900, lavender on Dusk
Light mode	Optional "Day" theme. Not drawn yet: derive it by swapping the grounds to #F7F4FB / #FFFFFF / #EEE8F7 with `--ink` text, keeping the same pastels at 100 % for fills

Still open:

- [ ] Final name and app icon
- [ ] Draw the Day theme screens before milestone 9
- [ ] Whether to localise (US 12-hour times, "OB/provider" wording) in v1 or later